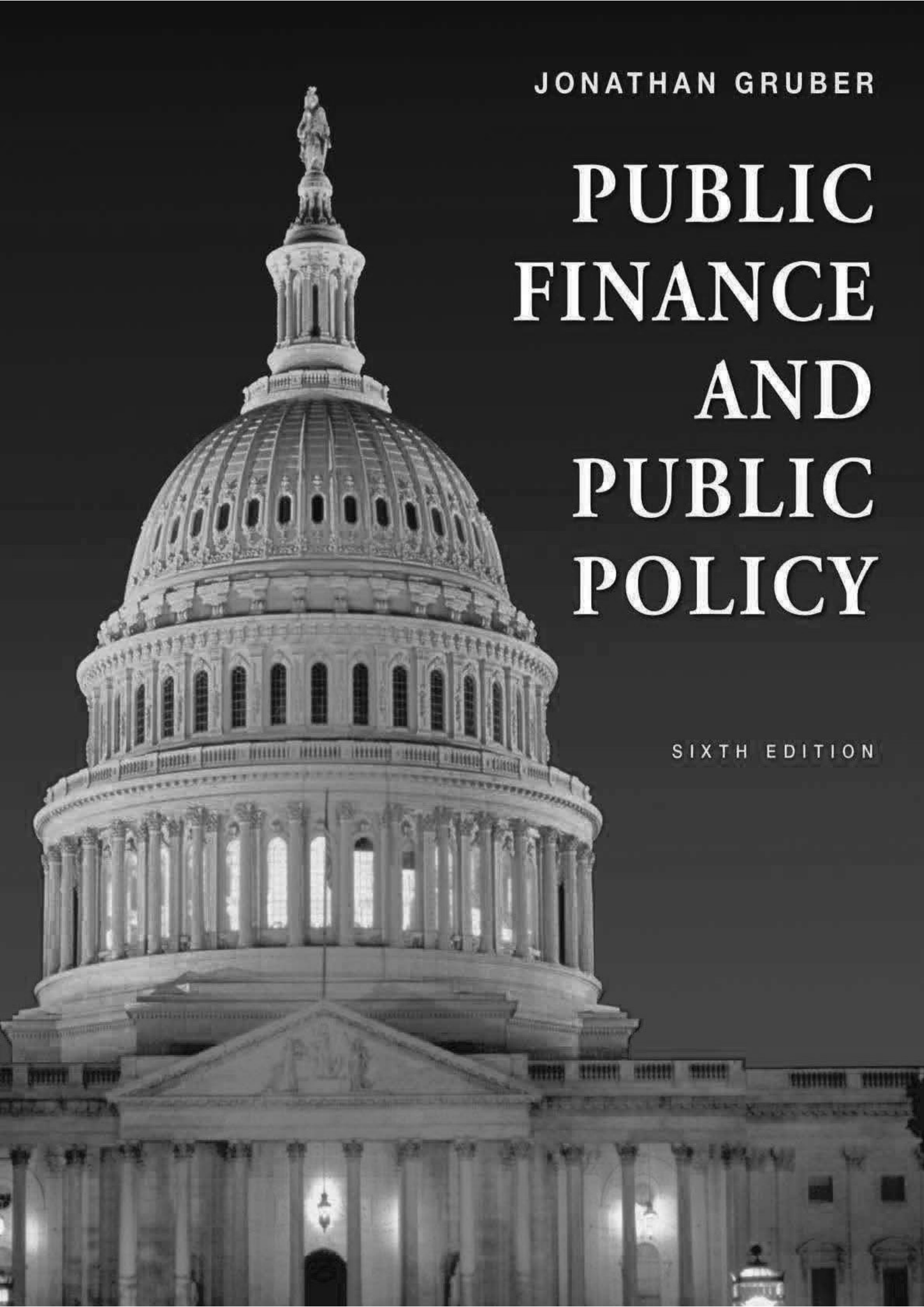
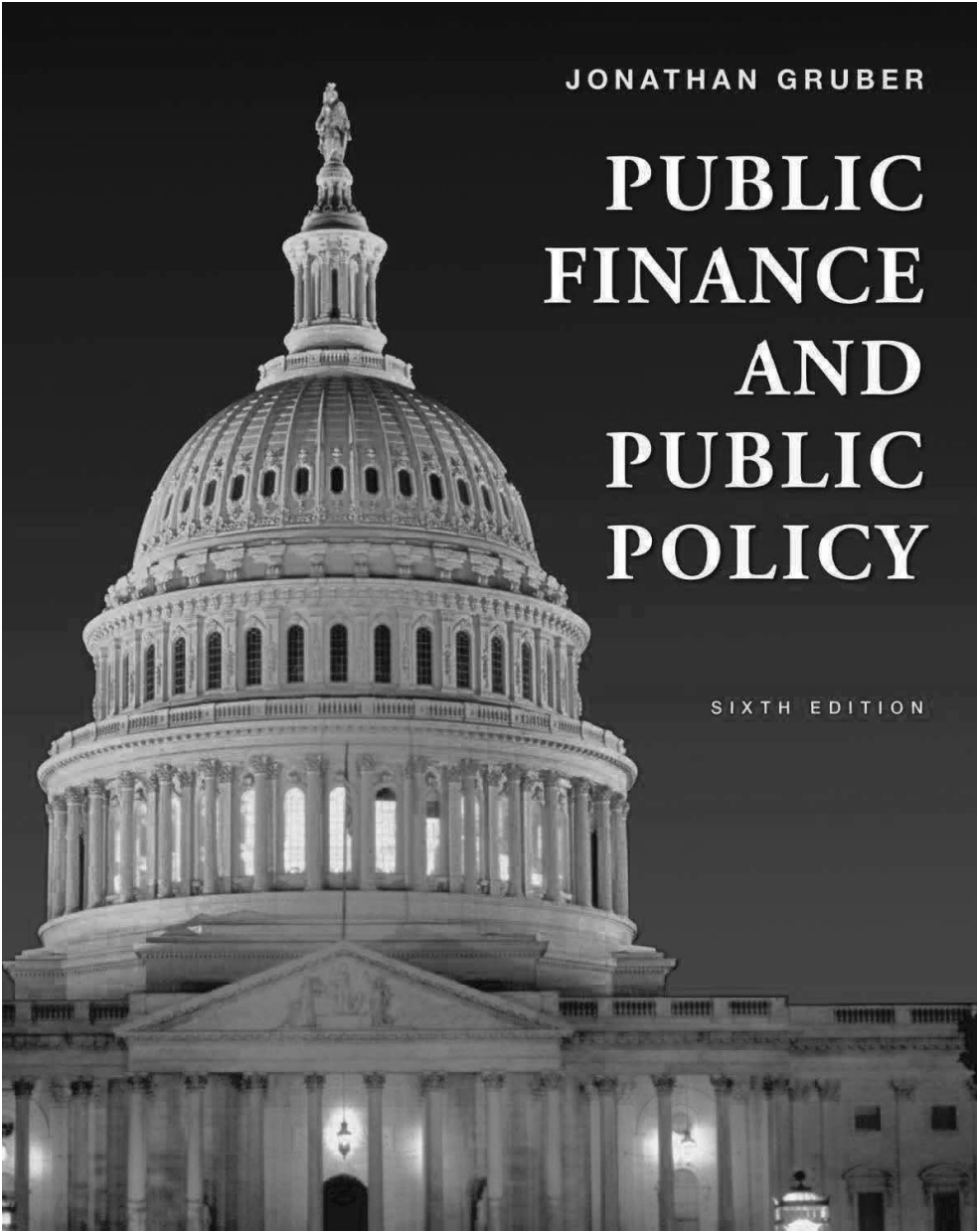




JONATHAN GRUBER

PUBLIC FINANCE AND PUBLIC POLICY

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CHAPTER 19 The Equity Implications of Taxation: Tax Incidence



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- [19.1 The Three Rules of Tax Incidence](#)
- [19.2 Tax Incidence Extensions](#)
- [19.3 General Equilibrium Tax Incidence](#)
- [19.4 The Incidence of Taxation in the United States](#)
- [19.5 Conclusion](#)
- [APPENDIX TO CHAPTER 19](#)
 - [The Mathematics of Tax Incidence](#)



Questions to keep in mind

- What factors determine who bears the burden of taxation?
- How does considering markets other than the one in which the tax is imposed affect the burden of taxation?
- What is the distribution of tax burdens in the United States?

In early 2002, with New Jersey facing a \$5.3 billion budget gap, Governor James McGreevey called for changes in the state's corporate tax system. The outdated system based corporate tax payments on the corporation's profits earned in the state of New Jersey, thus encouraging businesses to use accounting tricks to shift reported profits to their subsidiaries in other states. As a result, 30 of the state's 50 companies with the biggest payrolls each paid just \$200 annually in corporate taxes. McGreevey wanted to institute a 1% tax on corporate gross sales in New Jersey to ensure that all corporations would pay tax. Arthur Maurice, vice president of the NJ Business and Industry Association, objected: "Where are companies going to get the money to pay these taxes? They're going to cut jobs. It will be the people who work at these companies who will ultimately pay the price for this counterproductive tax."

A version of the tax reform was eventually enacted, leading one company, Federated Department Stores, to publicly announce layoffs that they attributed to the tax increase. Governor McGreevey responded angrily, claiming that these statements were just a cover for those wealthy corporate owners who would really bear the brunt of this new tax: "All that we're asking is that they pay their fair share: not a dollar more, not a dollar less. But when you have a CEO making \$1.5 million and upwards of \$14 million in stock options threatening people who are making \$25,000, that's what's wrong."¹

The fundamental disagreement between the governor and the business community concerned who would ultimately pay this new tax. The business community claimed workers would bear the burden, while the governor claimed the burden would be shouldered by wealthy companies and their executives.

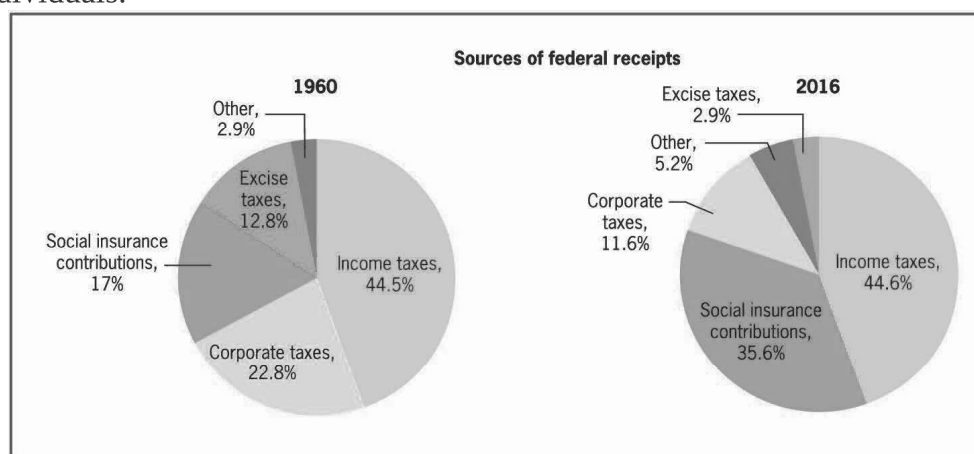
This debate focuses on the central question of **tax incidence**: Who bears the burden of a tax? A simple answer to this question would be that whoever sends the check to the government bears the tax. Yet such an answer ignores the fact that markets respond to taxes and that these responses must be taken into account to assess the ultimate burden, or *incidence*, of taxation.

tax incidence

Assessing which party (consumers or producers) bears the true burden of a tax.

To see the importance of this question, let's return to the facts on the distribution of federal taxation over time from [Chapter 1](#). [Figure 19-1](#) shows these facts again. In 1960, 23% of federal taxes was collected from corporations, and 61.5% was collected from individuals through income and payroll taxes. Today, 12% is collected from corporations, and 80% is collected through income and payroll taxes. Is this an equitable shift in the burden of taxation? The typical person's first reaction might be "No, it's not equitable at all! Corporations are rich, and individuals aren't, so this imbalance is not fair." Such an indignant reaction, however noble, is misplaced because corporations don't pay taxes; corporate taxes are paid by the individuals who own, work for, and buy from the corporations. When we study tax incidence, then, we are always comparing taxes collected from one set of people to taxes collected from another set. This comparison makes the issue of figuring out who bears

the burden of a tax less clear than if we view it simply as a matter of pitting rich corporations against individuals.



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FIGURE 19-1

The Sources of Federal Government Revenue (% of Total Receipts) • The federal government depends much less on corporate and excise taxes and much more on payroll taxes than it did in 1960. Its biggest source of revenue, then as now, is the individual income tax.

Data from: [Bureau of Economic Analysis \(2018\)](#), Table 3.1.

This chapter examines the equity implications of taxation. We begin with the *three rules of tax incidence* that guide our modeling of the distributional implications of taxation. We then turn to the study of *general equilibrium tax incidence*, the effect of taxes on one sector in a multisector world. Finally, we present empirical evidence on the burden of taxation in the United States over time.

19.1 The Three Rules of Tax Incidence

The goal of determining a tax's incidence is to assess who ultimately bears the burden of paying a tax. Economic tax incidence can be described by *three basic rules*. We describe these rules with reference to the incidence of a tax of a fixed amount on a specific commodity, or a *specific excise tax*. An alternative form of taxation of commodities is *ad valorem* taxes, a fixed percentage of the sales price (such as with state sales taxes). All of the lessons drawn here apply equally to both types of taxes; the major difference with ad valorem incidence analysis is that taxes shift the demand or supply curve proportionally (e.g., quantity rises by 10%) rather than by fixed amounts (e.g., quantity rises by five units).

The Statutory Burden of a Tax Does Not Describe Who Really Bears the Tax

The first and most important rule of tax incidence is that tax laws do not accurately identify who actually bears the burden of the tax. The **statutory incidence** of a tax is determined by who pays the tax to the government. For example, the statutory incidence of a tax paid by producers of gasoline is on those very producers. Statutory incidence, however, ignores the fact that markets react to taxation. This market reaction determines the **economic incidence** of a tax, the change in the resources available to any economic agent as a result of taxation. The economic incidence of any tax is the difference between the individual's available resources before and after the tax has been imposed.

statutory incidence

The burden of a tax borne by the party that sends the check to the government.

economic incidence

The burden of taxation measured by the change in the resources available to any economic agent as a result of taxation.

When a tax is imposed on producers in a competitive market, producers will raise prices to some extent to offset this tax burden, and the producers' income will not fall by the full amount of the tax. When a tax is imposed on consumers in a competitive market, the consumers will not be willing to pay as much for the taxed good, so prices will fall, offsetting to some extent the statutory tax burden on consumers. Technically, we can define the tax burden for consumers as

consumer tax burden

$$= (\text{post-tax price} - \text{pre-tax price}) + \text{per-unit tax payments by consumers}$$

For producers, the tax burden is

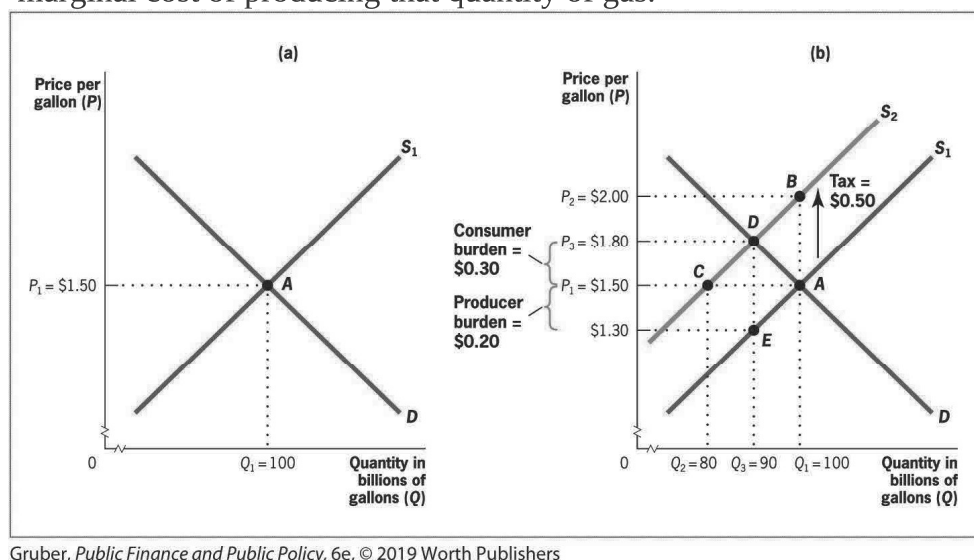
producer tax burden

$$= (\text{pre-tax price} - \text{post-tax price}) + \text{per-unit tax payments by producers}$$

For example, suppose that tomorrow the federal government levied a 50¢ per gallon tax on gasoline, to be paid by the producers. Will gas producers receive 50¢ less on each gallon they produce as a result of this tax?

To answer this question, we need to consider the impact of the gas tax on the market for gas, as shown in panel (a) of [Figure 19-2](#). The vertical axis in this graph shows the price per gallon of gas, and the horizontal axis shows billions of gallons of gas. Recall from [Chapter 2](#) that the supply curve shows the quantity that suppliers are willing to sell at any given price. In a competitive market, the supply curve is determined by the firm's marginal cost: the

producer will sell any units for which the market price is at or above its marginal cost of producing that unit. In Figure 19-2, the market is initially in equilibrium at point A: at the market price of **\$1.50 (P_1)**, producers will supply 100 billion gallons (**Q_1**) of gasoline. Producers are willing to supply 100 billion gallons at \$1.50 per gallon because \$1.50 is the producers' marginal cost of producing that quantity of gas.



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FIGURE 19-2

Statutory Burdens Are Not Real Burdens • Panel (a) shows the equilibrium in the gas market before taxation (point A). A 50¢ tax levied on gas producers (the statutory burden) in panel (b) leads to a decrease in supply from **S_1 to S_2** and to a 30¢ rise in the price of gas from **P_1 to P_3** (point D). The real burden of the tax is borne primarily by consumers, who pay 30¢ of the tax through higher prices, leaving producers to bear only 20¢ of the tax.

Panel (b) of Figure 19-2 shows the effects of imposing a tax of 50¢ per gallon of gas sold on the producers of gas. For these producers, this is equivalent to a 50¢ per gallon increase in marginal cost. Because firms must pay both their original marginal cost and the 50¢ tax, they now require a price that is 50¢ higher to produce each quantity. To supply the initial equilibrium quantity of 100 billion gallons after the tax is imposed, for example, firms would now require a price of **$P_2 = \$2.00$** (50¢ higher than the initial \$1.50 equilibrium price, at point B). Because the tax acts like an increase in marginal cost, the entire supply curve shifts upward by 50¢ from **S_1 to S_2** , and the supply of gas falls.

At the initial equilibrium price of \$1.50, there is now excess demand for gasoline. Consumers want the old amount of gasoline (100 billion gallons) at \$1.50, but with the new tax in place, producers are willing to supply only 80 billion gallons (point C). At \$1.50, there is a shortage of **Q_1 (point A) minus Q_2 (point C)**, or 20 billion gallons. Consumers therefore bid up the price as they compete for the smaller quantities of gas that are now available from producers. Prices continue to rise until the market arrives at a new equilibrium (point D) with a market price of **$\$1.80 (P_3)$** and a quantity of 90 billion gallons (**Q_3**). The market price is now 30¢ higher than it was before the tax was imposed.

Burden of the Tax on Consumers and Producers

The tax has two effects on the participants in the gas market. First, it has changed the market price that consumers pay and producers receive for a gallon of gas; this price has risen by 30¢

from \$1.50 to \$1.80. Second, producers must now send a check to the government for 50¢ for each gallon sold.

From the producers' perspective, the pain of the 50¢ tax is offset by the fact that the price the producers receive is 30¢ more than the initial equilibrium price. Thus, the producers have to pay only 20¢ of the tax, the portion that is not offset by the price increase.

From the consumers' perspective, they feel some of the pain of the tax because they pay 30¢ more per gallon. Even though consumers send no check to the government and producers send a 50¢ check to the government, consumers actually bear more of the tax (30¢ to the producers' 20¢). The price increase has transferred most of the tax burden from producers to consumers.

These burdens are illustrated in [Figure 19-2](#) by the segments labeled "Consumer burden" and "Producer burden." Using the formulas on [p. 587](#), we can compute the burdens on consumers and producers. The consumers' burden is

$$\begin{aligned} & \text{consumer tax burden} \\ & = (\text{post-tax price} - \text{pre-tax price}) + \text{per-unit tax payments by consumers} \\ & = P_3 - P_1 + 0 = \$1.80 - \$1.50 = \$0.30 \end{aligned}$$

The producers' burden is

$$\begin{aligned} & \text{producer tax burden} \\ & = (\text{pre-tax price} - \text{post-tax price}) + \text{per-unit tax payments by producers} \\ & = P_1 - P_3 + \$0.50 = \$1.50 - \$1.80 + \$0.50 = \$0.20 \end{aligned}$$

The key insight is that the burden on producers is not the 50¢ tax payment they make on each gallon but some lower number because some of the tax burden is borne by consumers in the form of a higher price. The sum of these burdens is \$0.50, the total **tax wedge** created by this tax, which is the difference between what consumers pay (\$1.80) and what producers receive net of tax (\$1.30, at point *E*) from a transaction.

tax wedge

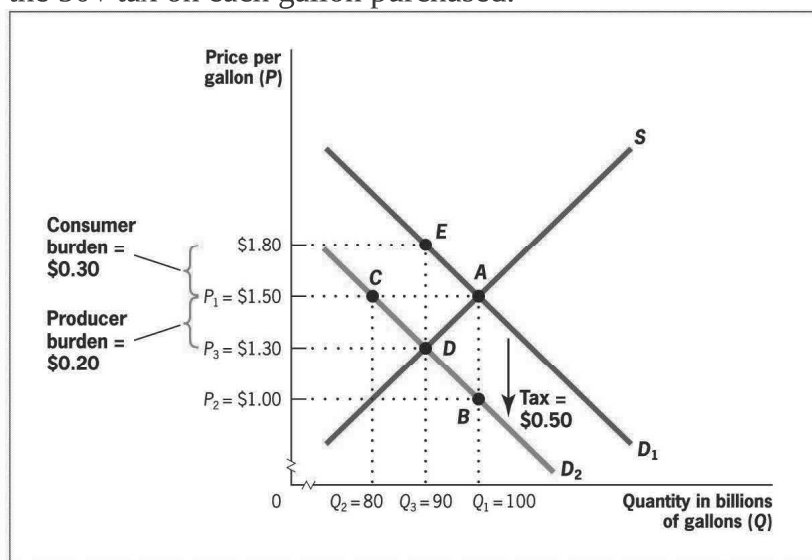
The difference between what consumers pay and what producers receive (net of tax) from a transaction.

The Side of the Market on Which the Tax Is Imposed Is Irrelevant to the Distribution of the Tax Burdens

The second rule of tax incidence is that the side of the market on which the tax is imposed is irrelevant to the distribution of the tax burdens: tax incidence is identical whether the tax is levied on producers or consumers.² In terms of the previous rule and [Figure 19-2](#), this rule means that whether the 50¢ tax is imposed on producers or consumers, consumers will always end up bearing 30¢ of the tax, and producers will end up bearing 20¢.

[Figure 19-3](#) considers the impact of a 50¢ per gallon tax on consumers of gas. In this case, the tax is collected from consumers at the pump when they pay for their gas rather than from producers, as in [Figure 19-2](#). Recall from [Chapter 2](#) that the demand curve represents

consumers' willingness to pay for any quantity of a good. Each point on the demand curve shows the quantity demanded for any market price encountered by consumers. With consumers having to pay a 50¢ tax in addition to the market price at every quantity, they are now willing to pay 50¢ less for each quantity. Thus, because the tax causes a reduction in consumers' willingness to pay (before adding in their tax payments), the entire demand curve shifts downward by 50¢, from D_1 to D_2 . Before the tax, consumers were willing to pay a price of $P_1 = \$1.50$ for the 100 billionth gallon of gas at point A. Now they are willing to pay only a price of $P_2 = \$1.00$ for the 100 billionth gallon (point B) because they also have to pay the 50¢ tax on each gallon purchased.



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FIGURE 19-3

The Side of the Market Is Irrelevant • A 50¢ tax levied on gas consumers (the statutory burden) leads to a decrease in demand from D_1 to D_2 and to a 20¢ fall in the price of gas from P_1 to P_3 (with the market moving from the pre-tax equilibrium at point A to the post-tax equilibrium at point D). The real burden of the tax is borne primarily by consumers, who pay the 50¢ tax to the government but receive an offsetting price reduction of only 20¢. Producers bear that 20¢ of the tax.

At the old market price of \$1.50, there is now an excess supply of gasoline: producers are willing to sell the old amount of gasoline (100 billion gallons, at point A), but consumers are willing to buy only 80 billion gallons at that price, at point C. There is an excess supply of gasoline of $Q_1 - Q_2 = 20$ billion gallons at the initial equilibrium price of \$1.50 after the demand curve shifts. Producers therefore lower their price to sell their excess supply until the price falls to $\$1.30$ (P_3) at point D, with an equilibrium quantity of 90 billion gallons (Q_3). The market price is now 20¢ lower than it was before the tax was imposed.

As in the previous example, this tax has two effects on the participants in the gas market. First, it has changed the market price that consumers pay and producers receive for a gallon of gas; this price has fallen by 20¢ from \$1.50 to \$1.30. Second, the consumer must now pay the government 50¢ for each gallon purchased. At the equilibrium price of \$1.30, adding the 50¢ tax yields a cost to consumers (price plus tax) of \$1.80 at point E.

From the consumers' perspective, the pain of the 50¢ check is offset by the 20¢ per gallon decline in the market price. From the producers' perspective, they are feeling some of the

pain of this tax since they are receiving 20¢ less per gallon. Even though producers send no check to the government, and consumers send a 50¢ check to the government, both parties bear some of the ultimate burden of the tax because the price decrease has transferred some of the tax burden from consumers to producers.

These burdens are illustrated in [Figure 19-3](#) by the segments labeled “Consumer burden” and “Producer burden.” Using our formulas, we can compute the burdens on consumers and producers:

consumer: $P_3 - P_1 + \$0.50 = \$1.30 - \$1.50 + \$0.50 = \$0.30$

producer: $P_1 - P_3 + 0 = \$1.50 - \$1.30 = \$0.20$

Once again, the sum of the burdens on consumers and producers, the difference between what consumers pay (\$1.80) and what producers receive (\$1.30), is the tax wedge of 50¢.

Note that *these tax burdens are identical to the burdens in the previous example*. Consumers now have to pay the 50¢ at the pump, but they are facing a lower price (\$1.30) to which they have to add that tax. Adding the two together, the consumer pays exactly the same amount (\$1.80, price plus tax) as in the previous case. Producers now don’t have to pay a tax, but they receive a lower price for their gas (\$1.30 instead of \$1.50), so they end up receiving the same amount (\$1.30) as well.

Gross Versus After-Tax Prices

While there is only one market price when a tax is imposed, there are two different prices that economists often track in these types of tax incidence models. The first is the **gross price**, the price paid by or received by the party not paying the tax to the government; it is the same as the price in the market. The second is the **after-tax price**, the price paid by or received by the party that is paying the tax to the government; it is either lower by the amount of the tax (if producers pay the tax) or higher by the amount of the tax (if consumers pay the tax).

gross price

The price in the market.

after-tax price

The gross price minus the amount of the tax (if producers pay the tax) or plus the amount of the tax (if consumers pay the tax).

When the gas tax is levied on producers, as shown in [Figure 19-2](#), the gross price paid by consumers is \$1.80, and the after-tax price received by producers is **$\$1.80 - \$0.50 = \$1.30$** . When the gas tax is levied on consumers, as in [Figure 19-3](#), the gross price received by producers is \$1.30, and the after-tax price paid by consumers is **$\$1.30 + \$0.50 = \$1.80$** . The after-tax price is equal to the gross price *plus* the tax wedge if the tax is on consumers, but is equal to the gross price *minus* the tax wedge if the tax is on producers.

Parties with Inelastic Supply or Demand Bear Taxes; Parties with Elastic Supply or Demand Avoid Them

In the previous example, we described a particular case in which consumers bear more of the burden of a tax than do producers. This is, however, only one of many possible outcomes. The incidence of taxation on producers and consumers is ultimately determined by the

elasticities of supply and demand on how responsive the quantity supplied or demanded is to price changes.

Perfectly Inelastic Demand

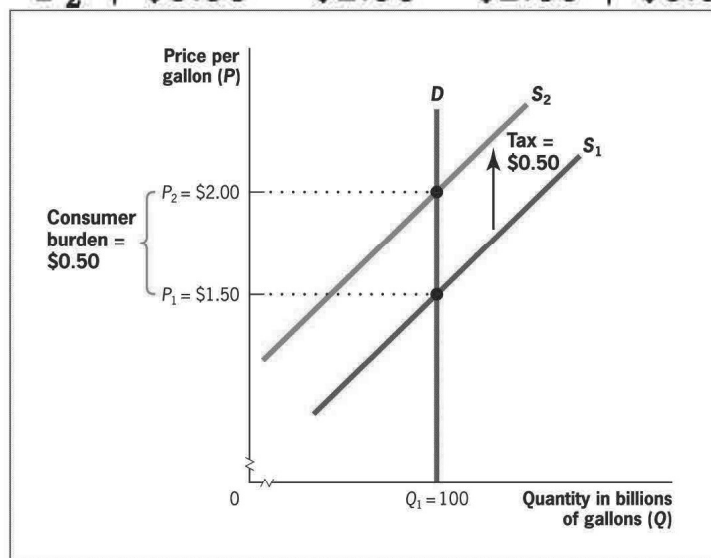
Consider again the case in which the 50¢ per gallon tax is levied on gasoline producers, but let's assume this time that consumers have a perfectly inelastic demand for gas, as shown in Figure 19-4. At initial equilibrium, the price for 100 billion gallons is P_1 (\$1.50). When the tax is levied on producers, they once again treat this as equivalent to a 50¢ rise in marginal cost, raising the price that they require to supply any quantity; supply falls, and the supply curve shifts from S_1 to S_2 . The new equilibrium market price is \$2.00 (P_2), a full 50¢ higher than the original price P_1 . When demand is perfectly inelastic, the tax burdens are

consumer burden

$$\begin{aligned} &= (\text{post-tax price} - \text{pre-tax price}) + \text{tax payments by consumers} \\ &= P_2 - P_1 = \$2.00 - \$1.50 = \$0.50 \end{aligned}$$

producer burden

$$\begin{aligned} &= (\text{pre-tax price} - \text{post-tax price}) + \text{tax payments by producers} \\ &= P_1 - P_2 + \$0.50 = \$1.50 - \$2.00 + \$0.50 = \$0 \end{aligned}$$



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FIGURE 19-4

Inelastic Factors Bear Taxes • A tax on producers of an inelastically demanded good is fully reflected in increased prices, so consumers bear the full tax.

When demand is perfectly inelastic, producers bear *none* of the tax and consumers bear *all* of the tax. This is called the **full shifting** of the tax onto consumers.

full shifting

When one party in a transaction bears all of the tax burden.

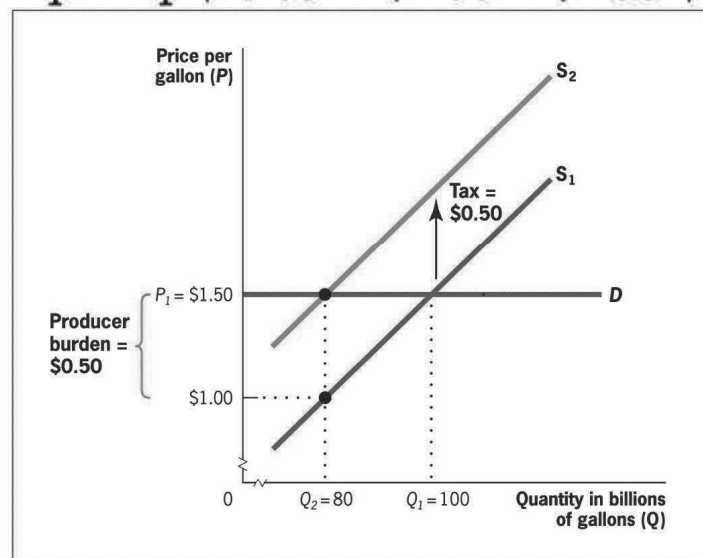
Perfectly Elastic Demand

Contrast that outcome with the case in which consumers' demand for gas is perfectly elastic,

as shown in Figure 19-5. Initially, the market is in equilibrium at $P_1 = \$1.50$ and $Q_1 = 100$ billion gallons. In this case, when a 50¢ tax causes the supply curve to shift from S_1 to S_2 , the equilibrium price remains at $P_1, \$1.50$, but the quantity falls to $Q_2, 80$ billion gallons. When demand is perfectly elastic, the tax burdens are therefore

consumer: $P_2 - P_1 = \$1.50 - \$1.50 = 0$

producer: $P_2 - P_1 + 0.50 = \$1.50 - \$1.50 + \$0.50 = \0.50



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FIGURE 19-5

Elastic Factors Avoid Taxes • A tax on producers of a perfectly elastically demanded good cannot be passed along to consumers through an increase in prices, so producers bear the full burden of the tax.

In this case, producers bear *all* of the tax, and consumers bear *none* of the tax.

General Case

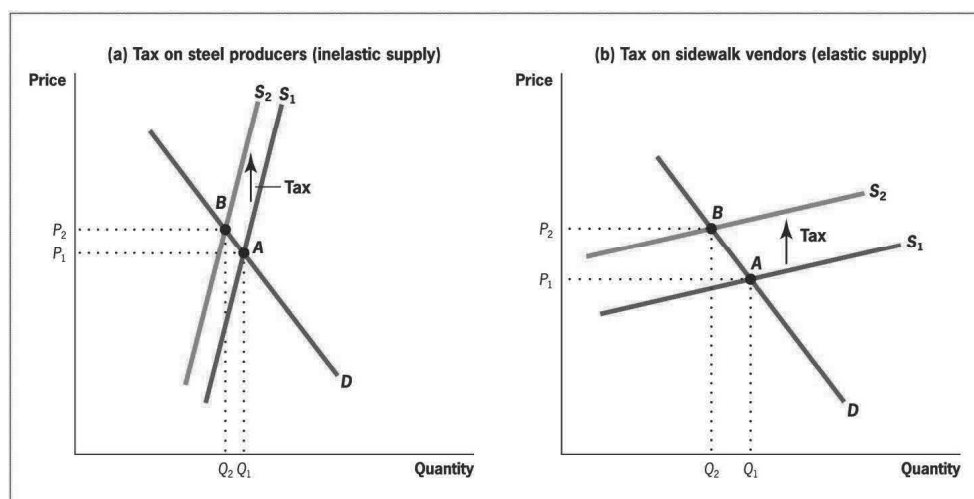
These extreme cases illustrate a general point about tax incidence: *parties with inelastic demand (or supply, as we'll soon see) bear taxes; parties with elastic demand (or supply) avoid them.* Demand for goods is more elastic (the price elasticity of demand is higher in absolute value) for goods with many substitutes. For example, the demand for fast food is fairly elastic because higher-quality restaurant meals or home cooking can be substituted for fast food fairly easily. Thus, if the government levied a tax on fast food, fast-food restaurants would find it difficult to raise prices in order to pass all of the tax onto fast-food consumers; if they did, individuals would substitute one of these alternatives for their fast food. Thus, because the demand for fast food is elastic, the producers (the restaurants) bear most of the burden of the tax.

For products with an inelastic demand, the burden of the tax is borne almost entirely by the consumer. For example, the demand for insulin is highly inelastic because it is essential to the health of diabetics. If the government taxes the producers of insulin, they can easily raise their price and completely shift most of the tax burden onto consumers because there are no substitutes available that allow consumers to leave this market due to a higher price.

Supply Elasticities

Supply elasticity also affects how the tax burden is distributed. Supply curves are more elastic when suppliers have more alternative uses to which their resources can be put. In the short run, a steel manufacturer has fairly inelastic supply; having invested in the steel plant and expensive machinery to produce steel, there are few alternative choices for production. The plant cannot easily convert from making steel to making plastic pipes or wood furniture. So the supply curve for steel will be fairly inelastic (vertical). The supply of sales from sidewalk vendors (of items such as watches, handbags, scarves, and so on) in New York City, in contrast, is very elastic. Because the individuals selling these goods have a very low investment in that particular business, if it is taxed, they can easily move to other activities, such as working in a store selling the same items. So the supply curve for sidewalk vendor sales will be very elastic (horizontal).

Compare the incidence of a tax on steel (levied on steel producers) to the incidence of a tax on sidewalk vendors (levied on the vendors) for any given demand curve (assuming that the demand curve is neither perfectly elastic nor inelastic). Panel (a) of Figure 19-6 shows the impact of a tax on steel producers. The steel market is initially in equilibrium at point A. The steel company can reduce the amount of steel it produces only slightly because it is committed to a level of production by its fixed capital investment. As a result, even when the steel company is paying 50¢ to the government for each unit of steel produced, it still wants to produce almost the same amount. Overall, the steel company's supply curve shifts upward from S_1 to S_2 . Price rises only slightly from P_1 to P_2 , and quantity of steel sold falls only from Q_1 to Q_2 ; the new equilibrium is at point B. Because the price rise is very small, it does not much offset the tax that the steel company must pay. The steel company therefore bears most of the tax, and consumers of steel bear very little (they don't pay a much higher price).



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FIGURE 19-6

Elasticity of Supply Also Matters • A tax on producers of an inelastically supplied good, as in panel (a), leads to a very small rise in prices, so producers bear most of the burden of the tax. An equal-sized tax on producers of an elastically supplied good, as in panel (b), leads to a large rise in prices, so producers bear little of the burden of the tax (and consumers bear most of the burden).

Panel (b) of Figure 19-6 shows the impact of an equal-sized tax on sidewalk vendors. These vendors are very sensitive to the costs of production in their production decisions, leading to the very elastic supply curve. They are initially willing to provide a quantity Q_1 of

goods at a price of P_1 . If the government makes them pay 50¢ per good they sell, then many vendors will move out of the sidewalk-vending business into some more lucrative line of work. The supply curve therefore shifts from S_1 to S_2 , with prices rising from P_1 to P_2 , and the quantity of goods sold falling from Q_1 to Q_2 (at point B). The large increase in price in the sidewalk vendors' market greatly offsets the taxes the vendors have to pay, so they bear little of the burden of the tax. Consumers of goods sold by sidewalk vendors will see much higher prices for these goods, however, so they will bear most of the tax.

Thus, the same principles hold for supply as for demand elasticities; elastic factors avoid taxes, while inelastic factors bear them.

In the appendix to this chapter, we develop the mathematical tax incidence formulas that formalize this intuition.

Reminder: Tax Incidence Is About Prices, Not Quantities

When the demand for gas is perfectly elastic, as in [Figure 19-5](#), we claimed that consumers bore none of the burden of taxation, and yet the quantity of gas consumed fell dramatically. Doesn't this decrease in consumption make consumers worse off? And if so, shouldn't that be taken into account when determining tax incidence?

The answer to both questions is "no" because, at both the old and new equilibria, consumers in this case are indifferent between buying the gas and spending their money elsewhere. Each point on a demand curve represents consumers' willingness to pay for a good. That willingness to pay reflects the value of the next best alternative use of their budget. If the demand curve for gas is perfectly elastic, consumers are truly indifferent, at the market price, between consuming gas and consuming some other good. So if they have to shift to buying more of another good and less gas, they are no worse off.

More generally, when we analyze tax incidence, we ignore changes in quantities and only focus on the changes in prices paid by consumers and suppliers. This assumption makes tax incidence analysis simpler.³

19.2 Tax Incidence Extensions

Section 19.1 presented the fundamental rules that will guide tax incidence analysis throughout the rest of this book. To recap:

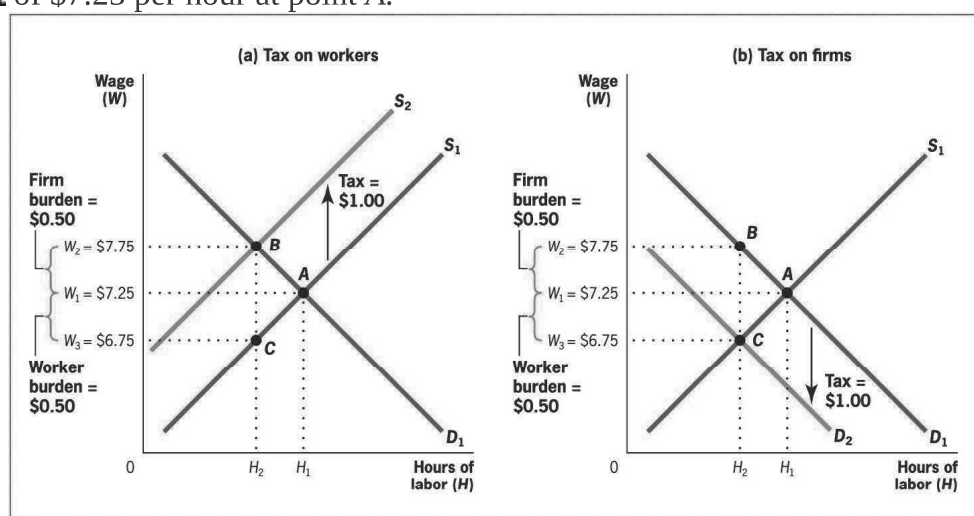
- The statutory burden of a tax does not describe who really bears the tax.
- The side of the market on which the tax is imposed is irrelevant to the distribution of tax burdens.
- Parties with inelastic supply or demand bear taxes; parties with elastic supply or demand avoid them.

In this section, we apply these rules to cases different from those previously considered, including taxes on factors of production, taxes in markets with imperfect competition, and accounting for (tax-financed) expenditures in tax incidence analysis. As we will see throughout the remainder of this book, the three basic rules of tax incidence are largely all we need to know to understand more complicated cases and issues in taxation.

Tax Incidence in Factor Markets

Our discussion thus far has focused on taxes that are levied in the goods markets, such as the markets for gas or fast food. Many taxes, however, are levied in factor markets, such as the market for labor. The analysis of tax incidence in factor markets is identical to that in goods markets; the only difference is that consumers of the factors are the firms (they demand factors such as labor) and producers of factors are individuals (who provide factors such as labor).

Consider, for example, the market for labor shown in panel (a) of Figure 19-7. Hours of labor supplied in the market are shown on the x axis; the market wage is on the y axis. There is a downward-sloping demand for labor from firms (D_1) and an upward-sloping supply of labor from individuals (S_1). The market is initially in equilibrium, before taxes, with a wage W_1 of \$7.25 per hour at point A.



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FIGURE 19-7

Incidence Analysis Is the Same in Factor Markets • These figures show the market for labor where firms are the consumers and workers are the producers of hours worked at a wage rate W . A \$1.00 tax per hour worked that is levied on workers,

shown in panel (a), leads the supply curve to rise from S_1 to S_2 and the wage to rise from its initial equilibrium value of \$7.25 (point A) to a higher value of \$7.75 (point B). A tax of \$1.00 per hour worked that is levied on firms, shown in panel (b), leads the demand curve to fall from D_1 to D_2 and the wage to fall from \$7.25 to \$6.75 at point C. Thus, regardless of who pays the tax, workers and firms each have a burden of 50¢ per hour.

Suppose that the government levies a payroll tax of \$1 per hour on all workers. This tax lowers the return to work by \$1 at every amount of labor. As a result, individuals require a \$1 rise in wages to supply any amount of labor and the supply shifts up from S_1 to S_2 in panel (a) of [Figure 19-7](#). With demand remaining at its original level, this shift results in a higher market equilibrium wage of \$7.75 (W_2) at point B. The incidence of the tax is shared by workers (suppliers) and firms (demanders) according to the elasticities of demand and supply. If these elasticities are equal, the burden is shared equally: the wage will rise to \$7.75 per hour, and workers will take home \$6.75 per hour after paying their \$1 per hour tax. The firms and the workers each bear 50¢ of the \$1.00 tax, split as indicated on the vertical axis. The firms pay a 50¢ higher wage (\$7.75), and the workers receive a 50¢ higher wage (\$7.75), but because they must pay \$1 an hour in tax, they receive 50¢ less in after-tax wage (\$6.75). The gross wage in the market has risen to \$7.75, but the after-tax wage of workers has fallen to \$6.75.

According to the second rule of tax incidence, what matters for the burdens on workers and employers from this tax is the total tax wedge and the elasticities of supply and demand, not who sends the check to the government. Panel (b) of [Figure 19-7](#) shows the effect on the labor market if the payroll tax in our example were instead paid only by firms and not by workers. In that case, the supply curve would remain at S_1 , and, because the tax on consumers (the firms) acts like an increase in the price of labor, the demand for labor would fall, and the demand curve would shift inward to D_2 . Market wages would fall by \$1.00 from \$7.75 to \$6.75, the new equilibrium (point C), and the burdens of taxation would be unchanged. Firms bear the same 50¢ burden as before; rather than paying a 50¢ higher wage, however, they now pay a wage (\$6.75) that is 50¢ lower than the initial equilibrium wage of \$7.25. In addition, firms now must send a \$1 check to the government, so in effect they are paying a wage of \$7.75.

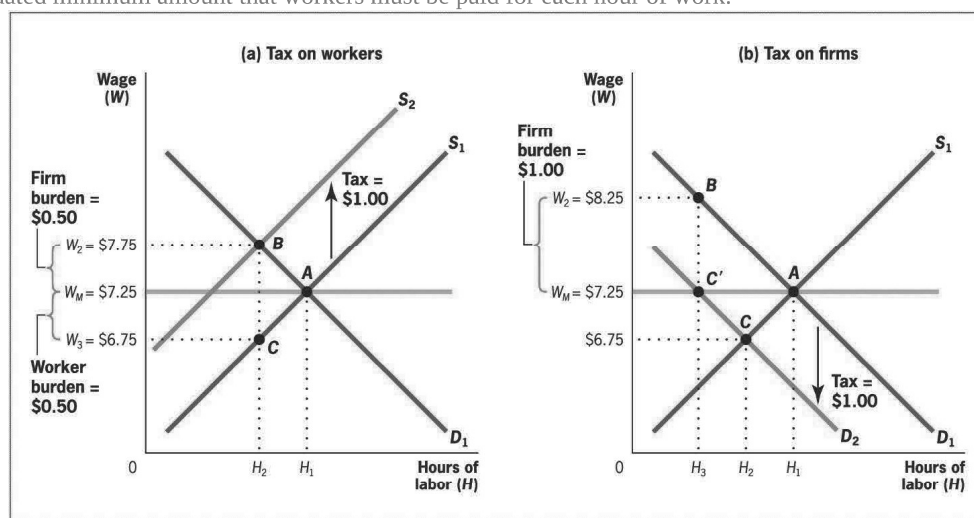
Workers see the same 50¢ burden; rather than receiving a 50¢ higher wage and sending a \$1 check to the government, however, they now receive a 50¢ lower wage (\$6.75). The gross wage in the market has fallen to \$6.75, but the after-tax wage paid by firms is \$7.75.

The tax incidence analysis of a payroll tax shows that it makes no difference that the Social Security payroll tax is levied half on workers and half on firms rather than being levied 100% on workers or on firms. The second rule of tax incidence tells us that what matters for determining the burden of the Social Security tax is the total size of the tax (the total tax wedge), not how the tax is distributed across demanders and producers.⁴

Impediments to Wage Adjustment

These conclusions from comparing panels (a) and (b) in [Figure 19-7](#) will not be correct, however, if anything impedes the free adjustment of wages in the labor market. One such impediment is a **minimum wage**, a mandated minimum amount that virtually all workers must be paid for each hour of work. The current U.S. minimum wage is \$7.25 per hour. Panels (a) and (b) of [Figure 19-8](#) show how the analysis of [Figure 19-7](#) changes when a

minimum wage, W_M , is introduced at the initial equilibrium wage of \$7.25.
 minimum wage
 Legally mandated minimum amount that workers must be paid for each hour of work.



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FIGURE 19-8

Incidence Analysis in Factor Markets Differs with a Minimum Wage • The analysis here is the same as in Figure 19-7, with the addition of the constraint that the wage cannot fall below \$7.25 per hour. If the payroll tax is levied on workers, as shown in panel (a), this constraint has no effect: the wage rises to \$7.75, as in Figure 19-7, and workers and firms equally share the burden of the tax. If the payroll tax is levied on firms, as shown in panel (b), however, the firms cannot lower the wage to the desired \$6.75 per hour, so the firms bear the full amount of the tax.

Panel (a) shows the case where the payroll tax is levied on workers. In that case, the minimum wage has no effect on incidence analysis: when the tax is implemented, wages rise to \$7.75 at point B (firms pay 50¢ in higher wages), workers send \$1 to the government in taxes, and they earn an after-tax wage of \$6.75.

Panel (b) shows the case where the payroll tax is levied on firms. In that case, the supply curve remains at S_1 , and the demand curve shifts to D_2 . However, the firms cannot lower the wage to the desired level of \$6.75 (point C) because this is an illegally low wage. The wage must stay at \$7.25, and the firm must bear all of the burden. The workers' wage doesn't fall from \$7.25, and the firm sends the check to the government, so the firm bears the full incidence of the payroll tax and pays an after-tax wage of \$8.25. The quantity of labor in the market falls to H_3 , where the new demand curve D_2 intersects the minimum wage line (point C').

When there are barriers to reaching the competitive market equilibrium (as in this minimum wage example), the side of the market on which the tax is levied can matter. A number of potential barriers, ranging from the minimum wage to workplace norms, do not allow employers to explicitly cut workers' wages. Such rigidities are often not present in output markets. For this reason, the party on whom the tax is levied may matter more in input than in output markets.

Tax Incidence in Imperfectly Competitive Markets

Our analysis thus far has focused on perfectly competitive markets in which there are a large number of demanders and suppliers consuming very similar goods. Very few markets are perfectly competitive in practice. Unfortunately, modeling *imperfectly competitive* markets is

an imperfect science. In this section, we discuss the one case that we can model clearly, a **monopoly market**, which is a market that has only one supplier of a good.

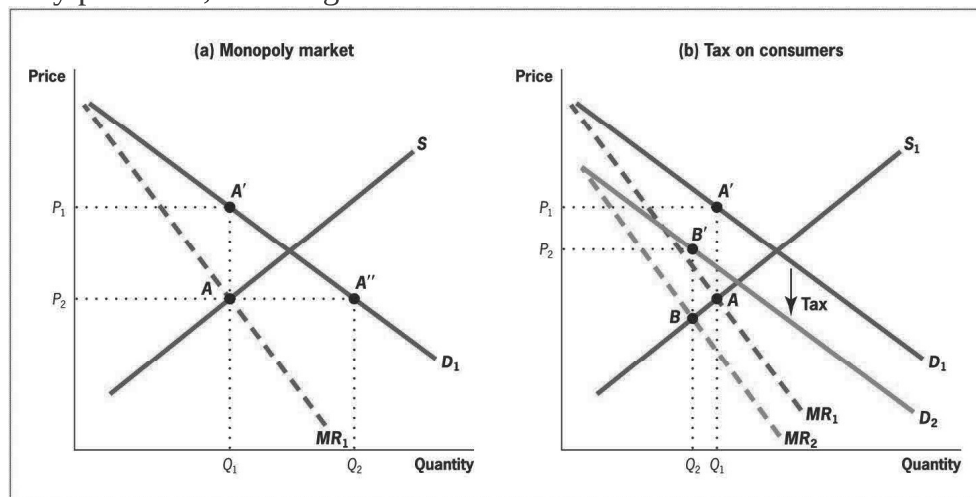
monopoly markets

Markets in which there is only one supplier of a good.

Background: Equilibrium in Monopoly Markets

Monopolists maximize profits just as competitive firms do: they produce a good or service until the marginal cost of the next unit produced equals the marginal revenues earned on that unit. For competitive firms, the marginal revenue earned on the next unit is the market price, so they set marginal cost equal to price. Because monopolists are *price makers*, not *price takers*, however, marginal revenue is *not* a price determined in the market but a price chosen by the monopolist. (In our analysis of tax incidence in monopoly markets, we assume that monopolists charge one price to all consumers; we do not consider the case of *price-discriminating monopolists*.)

Panel (a) of Figure 19-9 shows the determination of equilibrium in monopoly markets. The monopoly seller has an upward-sloping marginal cost curve and faces a downward-sloping demand curve. For a monopolist, there are two aspects to the decision about whether to produce and sell the next unit. The first is the price that the monopolist will earn on the next unit. The second is that in order to sell the next unit, the (non-price discriminating) monopolist must lower the price because he or she faces a downward-sloping demand curve. Consumers will buy another unit only if the market price is less than it was at the previous quantity. However, because the monopolist charges only one price to all customers, he or she must lower the price on all previous units for sale as well. Thus, monopolists face a trade-off as price makers: additional sales at a given price will increase revenue, but they will also force the monopolist to lower prices on all existing units to achieve equilibrium at the new higher quantity produced, lowering revenue.



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FIGURE 19-9

Tax Incidence in Monopoly Markets • Panel (a) shows the equilibrium in a monopoly market. The monopolist sets quantity produced where the marginal revenue curve intersects the supply curve (at Q_1) and then sets the price using the demand curve for that quantity (at P_1). When a tax is imposed on consumers in this market, as in panel (b), the demand curve shifts downward from D_1 to D_2 , leading the marginal revenue curve to also shift downward from MR_1 to MR_2 . The new equilibrium quantity is Q_2 , with a new price of P_2 .

The result of this pricing decision is that the monopolist's *marginal revenue curve* is the

line MR_1 in panel (a) of Figure 19-9, which lies everywhere below the demand curve D_1 . The marginal revenue that the monopolist gets from additional sales is below the consumers' willingness to pay for the given unit (the demand curve) because it incorporates the negative effect of lowering prices on all other units. In our example, the monopolist chooses to produce the quantity Q_1 , the quantity at which marginal revenue equals marginal cost at point A. As measured on the demand curve D_1 , consumers are willing to pay price P_1 for quantity Q_1 . Thus, the monopolist produces Q_1 and charges price P_1 . Note that even though the monopolist sets the price, the demand curve still must be respected. The monopolist cannot, for example, produce quantity Q_1 and charge price P_2 because at that price consumers would demand Q_2 far more than Q_1 (where the monopolist's marginal revenue equals marginal costs). The mathematics of the monopolist's profit maximization and its implications for tax incidence are developed in the appendix.

Taxation in Monopoly Markets

Suppose that the government imposes a tax on consumers in a monopoly market, as shown in panel (b) of Figure 19-9. This tax causes consumers to be less willing to pay a given market price for the monopolist's good, shifting the demand curve to D_2 . This reduction in willingness to pay leads to an associated shift in the marginal revenue curve to MR_2 . The new equilibrium (at the intersection of MR_2 and the marginal cost curve) is at a lower quantity Q_2 , and the new lower price is P_2 . The monopolist bears some of the tax, just as the competitive firm does: even though consumers are paying the tax to the government, the monopolist receives a lower price (P_2) in the market, so he or she shares the tax burden. The three rules of tax incidence therefore apply in monopoly markets as well. In this case, as we show in the appendix to this chapter, the side of the market on which the tax is imposed remains irrelevant: even though the monopolist has market power, a tax on either side of the market results in the same sharing of the tax burden. So monopolists cannot "exploit their market power" to avoid the rules of tax incidence.

Tax Incidence in Oligopolies

Very few markets are perfectly competitive or monopolistic. Most markets operate somewhere in between, with some degree of imperfect competition. **Oligopoly markets** are markets in which firms have some market power in setting prices but not as much as a monopolist. Although there are widely accepted models of how competitive and monopolistic markets work, there is much less consensus on models for oligopolistic markets. As a result, economists tend to assume that the same rules of tax incidence apply in these markets as well, but there is more work to do to understand the burden of taxes in oligopoly markets.

oligopoly markets

Markets in which firms have some market power in setting prices but not as much as a monopolist.

Balanced Budget Tax Incidence

The model we have used thus far to examine the burden of taxation in perfectly competitive markets and monopolies has focused on the incidence of the tax alone. Taxes also raise revenues, however, and these revenues will ultimately be spent. Thus, a complete picture of tax incidence would consider not only who bears the tax but also who receives the *benefit* of the spending that is financed out of the tax revenues.

Consider, for example, the federal gas excise tax, a tax levied by the federal government on the sale of gasoline. Suppose that demand for gasoline is very inelastic (or supply is very

elastic), so that the price of gas to consumers rises by the full amount of this tax. At the same, however, 84% of the revenues collected by this 18.3¢ gas tax are spent on highway improvements and repairs through the highway trust fund.⁵

In this case, it would be wrong to say that drivers fully bear the tax: they do pay the tax, but they also reap some of the rewards in terms of better roads on which to drive. Tax incidence that takes into account the incidence of both the tax and the benefit is called **balanced budget incidence**.

balanced budget incidence

Tax incidence analysis that accounts for both the tax and the benefits it brings.

It is often difficult, however, to trace the spending associated with a given tax increase. Thus, we typically ignore the spending side when we do tax incidence analysis. It is important to remember, however, that in reality the full burden of a tax policy will depend on the distribution of both the tax payments and the spending of the associated revenues.

19.3 General Equilibrium Tax Incidence

The model of tax incidence that we have used thus far is what economists call a **partial equilibrium tax incidence** model, which considers the impact of a tax on a market in isolation. In reality, however, a given market does not exist in isolation; it is integrated with a number of other markets. To study the effects on related markets of a tax imposed on one market, economists use the model of **general equilibrium tax incidence**.

partial equilibrium tax incidence

Analysis that considers the impact of a tax on a market in isolation.

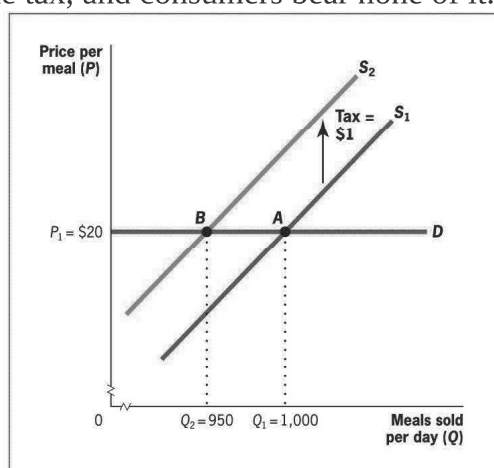
general equilibrium tax incidence

Analysis that considers the effects on related markets of a tax imposed on one market.

Effects of a Restaurant Tax: A General Equilibrium Example

Many travel guides consider Bozeman, Montana, to be among the most beautiful small towns in America. Suppose that Bozeman were to announce tomorrow that it was levying a tax of \$1 on all restaurant meals in that town. The demand for restaurant meals in Bozeman is fairly elastic because there are many substitutes, such as cooking at home or going to a restaurant in a nearby town. For ease, let's suppose that the demand for Bozeman restaurant meals is perfectly elastic.

The effect of the restaurant meal tax under this assumption is illustrated in Figure 19-10. The restaurant meal market in Bozeman is initially in equilibrium at point A: at a price of \$20 per restaurant meal (P_1), 1,000 restaurant meals are sold per day in the town (Q_1). The meal tax acts like an increase in the restaurants' marginal costs and shifts the supply curve inward from S_1 to S_2 , with quantity falling to 950 meals sold per day (Q_2). Note that this tax has no effect on the price charged by restaurants. Because demand is perfectly elastic, any increase in price to consumers would drive all business away. Thus, the restaurant bears the entire burden of the tax, and consumers bear none of it.



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FIGURE 19-10

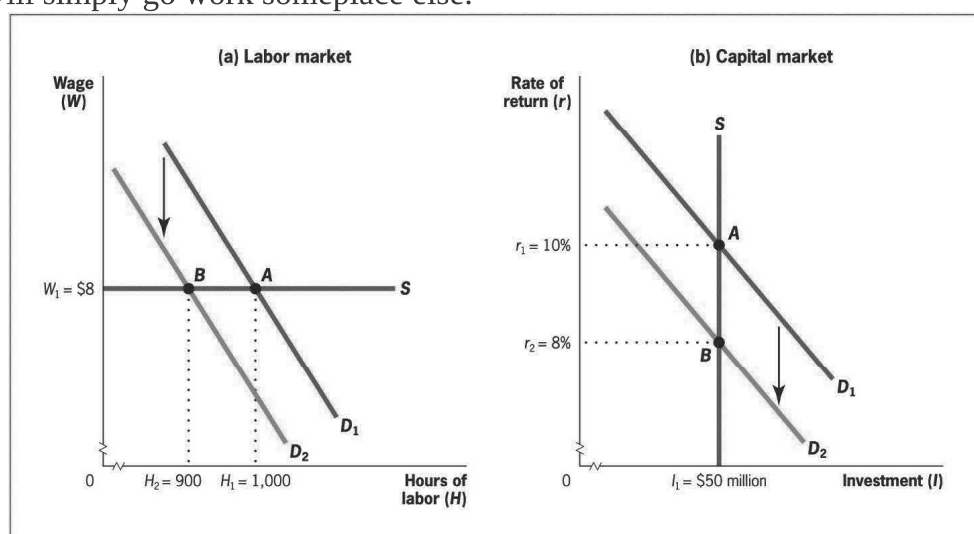
The Incidence of a Tax on Bozeman Restaurants • The demand for restaurants in Bozeman is perfectly elastic, so prices cannot increase when they are taxed; as a result of a \$1.00 tax on restaurant meals, the supply of meals falls from S_1 to S_2 , and the quantity of meals demanded and supplied falls to Q_2 (950). The price of a restaurant meal remains at \$20, so the restaurant, which is paying the tax, bears its full burden.

The story can't end there, however, for the simple reason that *restaurants cannot bear taxes*. As Nobel Prize-winning economist Milton Friedman once said in discussing energy taxes, "How do you 'burden' industry or 'tax' a factory? Do you squeeze it until it screams? Send it to jail? Only people can bear a 'burden' or pay a tax. An industry, a factory, or a utility can do neither."⁶

In the standard microeconomics model, firms are not self-functioning entities but are a technology for combining capital and labor to produce an output. In the context of our restaurant example, capital is best thought of as *financial* capital, the money that buys physical capital inputs, such as the building, the ovens, tables, and so on. By labor, we mean the hours of labor workers supply to the restaurant. When we say that the \$1 Bozeman meals tax is borne by restaurants, we mean that it is borne by the factors (labor and capital) that restaurants have organized to produce meals. To accurately identify who bears the burden of the meals tax, we need to move the analysis back one step and ask: In what proportions do these factors of production bear the restaurant tax?

General Equilibrium Tax Incidence

Consider first the market for labor employed by restaurants in Bozeman, shown in panel (a) of Figure 19-11. In this market, supply is likely to be very elastic because workers can always choose another job in Bozeman or go to work in a restaurant in a nearby town. Once again, for ease, let's assume that the labor supply available to restaurants in Bozeman is perfectly elastic. At the initial market equilibrium (point A), the wage is \$8 per hour (W_1), and the quantity of labor is 1,000 hours per year (H_1). When the new tax goes into effect, and restaurants bear its full burden (since the demand for restaurant meals is perfectly elastic), they will reduce their demand for workers. Each worker is worth less because a restaurant's willingness to pay for an hour of labor falls when it is taxed on the fruits of that labor (the meals).⁷ The demand curve in the Bozeman restaurant labor market shifts downward from D_1 to D_2 , but because labor supply is perfectly elastic, wages do not fall, and workers bear none of this tax. If restaurant owners try to pay their workers a wage lower than W_1 , the workers will simply go work someplace else.



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FIGURE 19-11

The Incidence of a Bozeman Restaurant Tax on Labor Versus Capital • If the burden of a tax on restaurants is borne by the restaurants, it must be borne by the factors of production used by the restaurants. In panel (a), the supply of labor to

restaurants in Bozeman is perfectly elastic, so when demand for labor falls to D_2 , it cannot be reflected in lower wages; the wage is unchanged and workers do not bear any of this tax. In panel (b), the supply of capital to restaurants in Bozeman is perfectly inelastic, so when demand for capital falls to D_2 , the rate of return to capital falls by the full amount of this tax to r_2 .

Now consider the market for capital in Bozeman restaurants, shown in panel (b) of [Figure 19-11](#). In the short run, having invested in a Bozeman restaurant, the capital owner is stuck, unable to pull out money that has already been spent on stoves, tables, and a building. In principle, the capital owner could resell goods, such as chairs, tables, and buildings, but in reality the owner would receive only a fraction of the purchase price. Thus, while the supply of capital isn't perfectly inelastic in reality, we assume for convenience in panel (b) of [Figure 19-11](#) that capital supply is perfectly inelastic.

The initial equilibrium in this capital market is at point A: the rate of return to capital is 10% (r_1) and the investment in restaurants is \$50 million. Because the tax is borne fully by the restaurants, it reduces their demand for capital just as it reduced their demand for labor. Capital is also worth less when the restaurant is taxed on the fruits of that capital (meals), so the restaurant will demand capital only from those who are willing to charge a lower rate of return. The lower demand is reflected in the fall in the demand curve from D_1 to D_2 ; the new equilibrium is at point B, with a lower rate of return of 8% (r_2). Because the supply of capital is inelastic in the short run, capital owners will bear the meals tax in the form of a lower return on their investment in the restaurant. Thus, when Bozeman levies a tax on restaurant meals, its incidence ultimately rests on the investors in Bozeman restaurants. This spillover of incidence to other markets is an example of general equilibrium tax incidence.

Issues to Consider in General Equilibrium Incidence Analysis

General equilibrium analysis is a game of “follow the tax burden”: the analysis does not stop with the effects on the market in which a tax is levied but expands to all related markets until it identifies the persons who ultimately bear the tax burden. This process raises three interesting issues that must be considered when using general equilibrium analysis.

Effect of Time Period on Tax Incidence: Short Run Versus Long Run

Under the assumptions we made, we have just shown the correct general equilibrium analysis of the tax burden of a restaurant meals tax in Bozeman. These assumptions are likely to be valid for a short-run (say, one-year) analysis of a restaurant tax. In the long run (e.g., more than ten years), however, these assumptions may no longer be valid. Over time, capital invested in Bozeman restaurants is no longer inelastic; investors can close or sell the restaurants, take their money, and invest it elsewhere. Indeed, in the long run, capital supply is likely to be perfectly elastic because there are many substitutes for investing in restaurants in Bozeman. In the long run, then, capital owners will not bear any of the tax.



Quick Hint What does it mean for capital supply to be elastic? Think of capital investments already made as irretrievable; that is why capital supply is inelastic in the short run. In the long run, however, restaurants need new infusions of capital to stay

afloat: equipment and furnishings need to be replaced, buildings need to be renovated, and so on. The elasticity of capital supply in the long run arises from the ability of investors to choose whether to reinvest in a firm. If there is a tax on the good produced by the firm, and this tax is passed on to capital investors in the form of a lower return, then they are less likely to reinvest in the restaurant.

The rule that the market participants with the most elastic supply or demand avoid taxes is *always* the guide to the game of “follow the tax burden.” We’ve just seen that in the long run, the supply of capital is perfectly elastic. The supply of labor was already elastic in the short run, and it should be even more elastic in the long run as workers with restaurant-specific skills can move out of Bozeman to other towns where the restaurants are not taxed and can pay higher wages as a result. This finding leaves us with a puzzle: If neither capital nor labor bears the tax in the long run, who does?

There is one more inelastic factor in the restaurant production process: the *land* that the restaurants are built on. This land is in fixed supply and is therefore inelastic by definition. When labor and capital avoid the tax, its burden falls squarely on the owners of the land in Bozeman. With workers unwilling to accept a lower wage to work at Bozeman restaurants and owners of capital unwilling to accept a lower return, the only way restaurants can stay open is if they pay lower rent on their land. Thus, factors that are always inelastically demanded or supplied in both the short and long run bear taxes in the long run.

Effect of Tax Scope on Tax Incidence

The incidence of a tax is also affected by the share of the product market to which it applies. Compare the incidence of the tax on restaurant meals in Bozeman with a tax imposed on all restaurant meals in Montana. While demand for meals in Bozeman may be very elastic, demand for all restaurant meals in Montana is less elastic because it is more costly to find substitutes. Individuals can still eat at home instead, but there aren’t easily available substitutes for dining away from home unless you live very near a neighboring state. Because demand is less elastic, restaurant consumers will bear some of the tax burden.

In the short run, the remainder of the tax burden not borne by consumers is shared once again between labor and capital. The incidence on labor has changed, however: workers are less mobile because they cannot move into other restaurant jobs as easily as they could when the restaurant tax was imposed only in Bozeman. (In that case, they could avoid the Bozeman tax burden by working at a restaurant in a neighboring town.) With the labor supply to the entire state restaurant industry less elastic than it was to just Bozeman restaurants, labor will bear some of the meals tax: restaurants will be able to pay their workers somewhat less in response to this tax because the workers have fewer employment alternatives. As before, capital is inelastic in the short run, so it will continue to bear much of the tax burden.

As a result, the short-run burden of a tax on all restaurants in Montana will be shared among consumers, workers, and capital, rather than being fully borne by capital. The exact proportions in which the burden will be shared depend on the exact elasticities of demand and supply in each of these markets. Once again, the goal of general equilibrium tax analysis is to follow the burden until it is distributed across all payers. When demands or supplies are not perfectly elastic or perfectly inelastic, the tax burden may be spread across many parties. The scope of the tax matters to incidence analysis because it determines which elasticities are relevant to the analysis: taxes that are broader based are harder to avoid than taxes that are narrower, so the response of producers and consumers to the tax will be smaller and more inelastic.

Spillovers Between Product Markets

While the discussion thus far has focused solely on the consumers and producers in the restaurant market, the incidence of a tax may extend beyond this market to other goods markets as well. When consumers of a good bear any of the tax on that good, this burden will affect their consumption of other goods by shifting their budget constraint. This spillover to other goods markets means that a tax in one market can have a burden or benefit on the consumers and producers in other markets, too.

Consider the tax on restaurant meals in Montana. This tax reduces demand for restaurant meals by raising the after-tax price to consumers. This higher after-tax price has three effects on other goods as well:

1. Consumers have lower incomes and may therefore purchase fewer units of all goods (the income effect).
2. Consumers may increase their consumption of goods and services (such as movies) that are substitutes for restaurant meals because they are now relatively cheaper than the taxed meals (the substitution effect).
3. Consumers may reduce their consumption of goods or services (such as valet parking services) that are complements to restaurant meals because they are consuming fewer restaurant meals (the complementary effect).

So, for example, a tax on restaurant meals could lower the demand for babysitters because incomes will be lower and individuals will go out to eat less frequently. On the other hand, the meals tax could raise the demand for concerts because individuals might attend a concert instead of eating out.

A complete general equilibrium tax analysis must account for effects in all other markets. Playing “follow the tax burden” means not only following it up and down within a market but also following it horizontally across markets and then vertically within those other markets. For example, if the tax on restaurants raises demand for movies (shifting out the demand curve), it leads to higher prices for moviegoers (so that movie consumers bear some of the tax on restaurants), higher wages for movie workers, and a higher return on capital in the movie sector. Indeed, a full general equilibrium analysis must also incorporate supply spillovers as well. As highlighted in the important article by [Bradford \(1978\)](#), a tax that shifts capital or labor out of any one jurisdiction will raise the supply of capital or labor to other jurisdictions, lowering the return to capital or labor elsewhere.

19.4 The Incidence of Taxation in the United States

The central role of tax fairness in debates over tax policy has motivated extensive analysis of tax incidence in the United States. The best-known study of tax incidence in the United States is the periodic analyses of the Congressional Budget Office (CBO), as updated for recent years by the Urban Institute's Tax Policy Center (TPC). In this section, we review the lessons from its analyses.

EMPIRICAL EVIDENCE

The Incidence of Taxation: Real-World Complications

While the three rules of tax incidence are clear, in reality, tax incidence can be a lot more complicated. In this box, we review two studies that illustrate those complications.

In the United States, excise taxes on goods such as gasoline, alcoholic beverages, and cigarettes vary widely across the states, and there are frequent changes within states over time in these taxes. This variation in taxation allows for quasi-experimental estimation of the impact of excise taxes on prices of the taxed goods. Analysts can compare the change in goods prices in the states raising their excise tax (the treatment group) relative to states not changing their excise tax (the control group), to measure the effect of each 1¢ rise in excise taxes on goods prices.

An excellent example is excise taxes on cigarettes. The excise tax on cigarettes varies widely across the U.S. states, from a low of 17¢ per pack in Missouri to a high of \$4.35 per pack in New York.

[Harding et al. \(2012\)](#) studied the impact of cigarette excise tax increases on cigarette prices over the 2006–2007 period. They made use of an innovative source of data for measuring prices: Neilson Homescan data. The Neilson company collects information on prices paid for many consumer products from a very large sample of individuals around the country, and the authors were able to obtain that detailed price information, including the prices paid for cigarettes. The authors do not just compare prices in high-tax versus low-tax states because this would be potentially biased by the fact that states may set their taxes as a function of voters' taste for smoking—tastes that also directly affect cigarette prices. So the fact that some Southern states have very low cigarette taxes and high smoking rates could reflect the causal effect of low taxes—or they could reflect a long-standing tradition of smoking in these tobacco-producing states, which leads politicians to keep cigarette taxes low.

To address this possible bias, Harding et al. used the quasi-experiment provided by tax changes in 12 states during their study period and studied how prices change when taxes are increased. Such tax increases are unlikely to be due to a sharp rise in the taste for smoking or other factors that drive pricing decisions. Thus, any changes in cigarette prices in those states, relative to similar states that did not raise taxes, should be due to the tax increase. The authors found that the vast majority of these tax increases, 85%, were passed forward to customers in the form of higher prices, suggesting fairly inelastic demand for cigarettes (consistent with the estimates discussed in [Chapter 6](#)).

Interestingly, however, they found that these price increases were not uniform throughout states that raise taxes; in particular, price increases were much lower near state borders. This is because customers in those areas can readily cross over to nearby low-tax states to buy cigarettes. Indeed, the authors showed a rise in cross-border shopping when taxes rose. As a result of this much more elastic demand, only 49% of tax increases were shifted to prices near state borders. More generally, the shifting of taxes to prices rose by 7.6% for each 1% increase in the distance to the state border. Thus, the average incidence of tax can mask considerable differences in underlying price elasticity and, therefore, the shifting of the tax increases to increases in prices.

Another complication to the rules of tax incidence can arise when there is differential tax evasion on different sides of the market. An example of this phenomenon is presented in [Kopczuk et al. \(2016\)](#). This paper studies the incidence of state diesel fuel taxes. An important feature of these taxes is that they are paid at different points in the supply chain in different states. In some states, the taxes are paid by retail outlets that sell diesel fuel. In other states, the taxes are paid by wholesalers who distribute diesel fuel to these retailers. But as we learned earlier, unless there are barriers to shifting tax increases to consumer prices, it shouldn't matter which supplier pays the tax—the incidence is determined by elasticities of supply and demand.

As Kopczuk et al. illustrated, however, this was no longer true once they recognized that retail outlets were more able to evade the tax (i.e., cheat the government) than were wholesale suppliers. This is because retail suppliers are more able to mislabel diesel fuel used for driving (which is subject to state taxes) as diesel fuel used for heating (which is not); such mislabeling is much harder at the wholesale level when fuel is being sold in bulk to retailers who sell both types of fuel. It is also because retailers are more able to simply underreport how much fuel was sold and, therefore, pay less in taxes compared to wholesale suppliers who are much more tightly monitored by the government and more likely to get caught if they underreport. If retail outlets can cheat and pay less tax, then there is less tax to be shifted to consumers, and prices will rise by less than for equivalent taxation of wholesalers.

The point at which taxes are paid changes over time and across states, so Kopczuk et al. were able to use a standard difference-in-difference estimation strategy to evaluate how the party paying the taxes impacted their pass-through to prices. The authors found that a higher percentage of the tax was passed through to increased prices when taxes were paid at the wholesale level. Because retailers cheated more—and, as a result, paid less tax—prices didn't rise by as much when taxes were levied at that level. Thus, in the case of diesel fuel taxes, whether the tax was levied on the wholesaler or the retailer affected how much of the tax was passed on to the consumer.

Another complication to our models in the real world is that markets are imperfectly competitive, with firms earning profits that should be competed away in perfect competition. As noted in [Section 9.2](#), we do not have strong predictions for tax incidence in oligopolistic industries.

A paper by [Ganapati et al. \(2018\)](#) explores the incidence of electricity costs (and therefore energy taxes) in oligopolistic industries. Of course, electricity costs might vary for reasons that are correlated with firm outcomes. To address this point, the authors used a “shift-share” analysis. That is, different parts of the country and different industries use electricity generated by a very different mix of energy inputs: electricity generated in some areas rely more on coal, while others rely more on natural gas, and still others more on oil. These energy prices often move independently of each other. For example, oil prices may rise while coal prices do not (due to an international oil embargo or a war in the Middle East, for example). In such cases, areas that generate electricity from oil will see prices go up, while those that rely on coal will not—providing a treatment/control comparison for analyzing pass-through. The authors also looked at industries that rely on different mixes of energy inputs in their production processes to measure the incidence of energy costs across industries. Using this variation across both areas and industries, they found that only about 70% of the costs of energy increases were passed on to consumers; the remainder came from firms in the form of lower profit rates.

A final real-world complication is that formal markets may compete with informal or illegal markets, significantly increasing consumer price sensitivity. This complication is explored in a novel study of the taxation of recreational marijuana in Washington state by [Hansen et al. \(2017\)](#). They studied the impact of shifting from a tax on suppliers that was collected at each stage of the production process to a tax that was collected solely at retail sales. They arrived at two key findings. First, taxing each stage of the production process led to an increase in “vertical integration,” with firms combining (inefficiently) to avoid taxes. Therefore, when the tax was changed, some of the benefits went to producers as they moved to a more efficient form of production. Second, consumers of marijuana are very price elastic—much more so than for goods such as cigarettes and alcohol—perhaps due to the possibility of substituting out of the legal market. As a result, the authors estimated that less than half of the incidence of marijuana taxes fell on consumers.

CBO/TPC Incidence Assumptions

The CBO and TPC analysis considers the incidence of the full set of taxes levied by the federal government. To do so, CBO/TPC must use the type of analysis we employed to guide their assumptions about the incidence of each type of taxation in the United States. Their key assumptions follow:

1. *Income taxes* are borne fully by the households that pay them.
2. *Payroll taxes* are borne fully by workers, regardless of whether these taxes are paid by the workers or by the firm.
3. *Excise taxes* are fully shifted to prices and so are borne by individuals in proportion to their consumption of the taxed item.
4. *Corporate taxes* are borne 20% by workers and 80% by the owners of capital; the latter is in proportion to each individual's capital income.

These assumptions are generally consistent with both theory and empirical evidence. For example, as we show in [Chapter 21](#), the low response of the labor supply of primary earners in the United States to taxation (inelastic supply) means that taxes on labor will be borne by workers. There is also substantial evidence that excise taxes on goods such as tobacco and alcohol are shifted forward to the prices charged to consumers, as discussed in the Empirical Evidence box.

The most difficult assumption to assess is the incidence of the corporate tax. As we'll discuss in [Chapter 24](#), it seems likely that at least some of the corporate tax is borne by consumers and workers rather than its full incidence being solely on capital owners. The

CBO/TPC traditionally assumed that the corporate tax was fully borne by the owners of capital, but more recently moved to the updated assumption noted above. Nevertheless, the available evidence is consistent with more than 20% of the burden being borne by workers. This finding suggests that the corporate tax is not as progressive as the CBO/TPC analysis indicates; this is because some of that tax is borne by lower-income workers (whereas the CBO/TPC assumes that the corporate tax is fully borne by higher-income owners of capital).

Results of CBO/TPC Incidence Analysis

The results of this CBO/TPC analysis are shown in [Tables 19-1](#) and [19-2](#); data through 2006 come from the CBO, while consistent data for 2017 are provided by TPC. [Table 19-1](#) shows the computed average tax rates (taxes paid relative to total income) by income group for the years 1979, 1985, 1990, 1995, 2000, 2006, 2011, 2014, and 2017. We show the results for all households and for the lowest and highest quintile. The top panel in [Table 19-1](#) shows the average tax rates for all taxes combined; the remaining panels show the average tax rates by type of tax.

TABLE 19-1 Effective Tax Rates

	1979	1985	1990	1995	2000	2006	2011	2014	2017
Total effective tax rate									
All households	22.2%	20.9%	21.5%	22.6%	23.0%	20.7%	18.6%	19.2%	20.2%
Bottom quintile	8.0%	9.8%	8.9%	6.3%	6.4%	4.3%	1.1%	3.1%	3.3%
Top quintile	27.5%	24.0%	25.1%	27.8%	28.0%	25.8%	23.8%	25.1%	26.0%
Effective income tax rate									
All households	11.0%	10.2%	10.1%	10.2%	11.8%	9.1%	10.1%	9.2%	10.0%
Bottom quintile	0.0%	0.5%	-1.0%	-4.4%	-4.6%	-6.6%	-6.0%	-4.5%	-4.9%
Top quintile	15.7%	14.0%	14.4%	15.5%	17.5%	14.1%	15.3%	15.1%	15.8%
Effective payroll tax rate									
All households	6.9%	7.9%	8.4%	8.5%	7.9%	7.5%	5.3%	7.0%	6.9%
Bottom quintile	5.3%	6.6%	7.3%	7.6%	8.2%	8.5%	4.8%	6.6%	7.1%
Top quintile	5.4%	6.5%	6.9%	7.2%	6.3%	5.8%	4.1%	5.9%	5.8%
Effective corporate tax rate									
All households	3.4%	1.8%	2.2%	2.8%	2.4%	3.4%	2.5%	2.9%	3.1%
Bottom quintile	1.1%	0.6%	0.6%	0.7%	0.5%	0.5%	0.4%	1.0%	1.1%
Top quintile	5.7%	2.8%	3.3%	4.4%	3.7%	5.4%	4.0%	4.0%	4.2%
Effective excise tax rate									
All households	1.0%	0.9%	0.9%	1.0%	0.9%	0.7%	0.7%	0.8%	0.7%
Bottom quintile	1.6%	2.2%	2.0%	2.4%	2.3%	1.9%	1.9%	2.1%	1.5%
Top quintile	0.7%	0.7%	0.6%	0.7%	0.5%	0.4%	0.4%	0.4%	0.5%

Data from: [Congressional Budget Office \(2007b\)](#) and [Tax Policy Center \(2018b\)](#).

TABLE 19-2 Top and Bottom Quintile's Share of Income and Tax Liabilities

	1979	1985	1990	1995	2000	2006	2012	2013	2014	2017
Top quintile										
Share of income	45.5%	48.6%	49.5%	50.2%	54.8%	55.7%	56.2%	54.5%	51.4%	52.6%
Share of tax liabilities	56.4%	55.8%	57.9%	61.9%	66.6%	69.3%	70.0%	68.0%	67.3%	67.6%
Bottom quintile										
Share of income	4.8%	4.6%	4.6%	4.0%	3.9%	3.9%	3.6%	3.6%	4.5%	4.0%
Share of tax liabilities	2.1%	2.3%	1.9%	1.3%	1.1%	0.8%	0.3%	0.3%	0.7%	0.6%

Top 1%										
Share of income	9.3%	11.5%	12.1%	12.5%	17.8%	18.8%	18.5%	17.8%	17.1%	15.5%
Share of tax liabilities	15.4%	14.8%	16.2%	20.1%	25.5%	28.3%	25.2%	27.0%	29.8%	25.4%

Data from: [Congressional Budget Office \(2007c\)](#) and [Tax Policy Center \(2018c\)](#).

This table shows the share of income and tax liabilities accruing to the top and bottom income quintiles over time.

Several interesting conclusions emerge:

1. The total average tax rate, combining all taxes across all households, declined from 22.2% in 1979 to 20.9% in 1985, then climbed to 23% in 2000, before falling until 2011, where it hit 18.6%. By 2017, the total average tax rate had risen again to 20.2%.
2. The total average tax rate on the bottom quintile rose from 8 to 9.8% from 1979 to 1985 and then fell throughout the rest of the period, reaching 4.3% by 2006 and falling sharply in the recession to only 1.1%. Since then, it has increased to 3.3% by 2017.
3. The average tax rate on the top quintile fell from 27.5% in 1979 to 24% in 1985, rose to 28% by 2000, and then fell back to 23.8% by 2011 before rising again in 2017 to 26%.
4. The bottom quintile pays much more in payroll taxes than in income taxes, and this disparity has been growing over time. In 1979, the bottom quintile had an average income tax rate of zero but an average payroll tax rate of 5.3%; by 2006, the average income tax was -6.6% (on net, the income tax system paid out more to lower-income individuals through tax credits, primarily the Earned Income Tax Credit, than it collected through tax payments), but the average payroll tax rate had risen to 8.5%. The payroll tax rate fell again in 2011 due both to the recession and to the temporary payroll tax holiday, but increased in 2017 to 7.1%. Indeed, more than two-thirds of all households paid more in payroll taxes than in income taxes in 2016.⁸
5. Average corporate tax rates are small relative to income and payroll tax rates and have fallen at both the top and bottom of the income distribution since 1979. The recent revised assumptions that allow some of the burden of the corporate tax to fall on workers has doubled the share borne by the bottom quintile in the most recent analysis.
6. Average excise tax rates are also small and have fallen for both the top and bottom of the income distribution, though they have fallen more steeply at the top. (This is due to a much larger decline in consumption of taxed goods, such as cigarettes, by higher-income groups than by lower-income groups.)⁹

The general conclusion from [Table 19-1](#) is that the United States has a fairly progressive tax system overall, although some elements (the income tax) are progressive while others (the payroll tax) are not. [Table 19-2](#) presents the share of tax liabilities paid by the top and bottom quintiles over time, and it compares this to their share of the income earned in the United States.¹⁰ The bottom quintile of taxpayers has always paid a very small share of taxes, and that share has fallen (from 2.1% of taxes paid to 0.7% of taxes paid) along with its share of national income (which has fallen from 4.8% of national income to 4% of national income). The top quintile of taxpayers has always paid the majority of taxes, and that share has risen (from 56.4% of taxes paid to 67.6% of taxes paid) along with its share of national income (from 45.5% of national income to 52.6% of national income). Today, the top quintile earns

more than half of all the income earned in the United States, and pays more than two-thirds of the taxes paid, under the CBO's incidence assumptions.

Of particular interest is the tax burden on the very richest members of society, which is also shown in [Table 19-2](#). The top 1% of the income distribution now controls 15.5% of pre-tax income, or almost one in every \$5 of income, a doubling since 1979. But the tax burden on this group has risen over time as well, increasing by almost two thirds since 1979.

Current Versus Lifetime Income Incidence

A final important distinction that must be drawn in incidence analysis is between **current tax incidence** and **lifetime tax incidence**. The CBO analysis just presented measures individual incomes on an annual basis. This approach can be misleading given the extensive mobility across income classes observed in the United States. Recent estimates show that between 25 and 40% of Americans change income quintiles within a one-year period, and roughly 56% change within a decade.¹¹

current tax incidence

The incidence of a tax in relation to an individual's current resources.

lifetime tax incidence

The incidence of a tax in relation to an individual's lifetime resources.

Closer to home, suppose that we were to assess the incidence of a tax on college textbooks. College students are typically living independently and would be measured as having fairly low incomes. Thus, such a tax would be viewed as highly regressive on an annual or current basis. Yet most of the students reading this book certainly hope that their incomes will be much higher later on in life as a result of their valuable college education. Indeed, the lifetime incomes of college graduates are, on average, more than twice as high as the lifetime incomes of those who do not attend college. Thus, on a lifetime basis, taxes on college textbooks are *progressive* because they tax those who earn high incomes on a lifetime basis.

The importance of this point is illustrated by [Poterba \(1989\)](#), who examined the incidence of alcohol, gasoline, and tobacco taxes. Poterba first divided people in 1984 by quintiles of income and observed that the share of income spent on gasoline and alcohol of the lowest quintile was five times the share of the highest quintile. The share of tobacco of the bottom quintile was nine times the share of the top quintile. Thus, it seems that taxes on such goods would be highly regressive.

Poterba then divided people by quintiles of *consumption*, which may be a better measure of lifetime resources. For example, low-income college students who know they will someday be rich will reflect that in their spending by borrowing from parents; similarly, rich elderly people who currently have low income but high savings will also have high consumption. Relating taxes paid to consumption rather than to current income, Poterba found that the tax burden is still regressive but much less so. In fact, the lowest four quintiles spend roughly the same share of their total expenditures on gasoline and alcohol, which is less than twice the highest quintile's share. For tobacco, the lowest quintile's share of total expenditures is only three times the share of the highest quintile.

19.5 Conclusion

Debates over tax incidence have been central to tax policy for decades. The “fairness” of any tax reform is one of the primary considerations in policy makers’ positions on tax policy. Therefore, it is crucial for public finance economists to have a deep understanding of who really bears the burden of taxation so that we can best inform these distributional debates over the fairness of a proposed or existing tax. The techniques developed in this chapter can provide that deep understanding and allow you to think more carefully through the distributional implications of any tax change.

HIGHLIGHTS

- Tax incidence is the study of the distribution of tax burdens across economic agents, and it takes into account not only who pays the tax to the government but also the effects of the tax on market prices.
- Tax incidence does not depend on who pays the taxes to the government or on whether taxes are levied on suppliers or demanders. In all cases, elastic parties avoid taxes and inelastic parties bear them.
- The incidence of taxes on factor markets is analyzed in the same way as taxes in goods markets, although impediments to price adjustment such as minimum wages can complicate the analysis. Likewise, the analysis of tax incidence in imperfectly competitive markets is similar to the analysis in competitive markets.
- General equilibrium tax incidence is a game of “follow the tax burden” from goods markets to their associated input markets and from one output market to another.
- Taxes are fairly progressive overall in the United States, although the overall progressivity of taxation is subject to key assumptions about tax incidence.
- Tax incidence can be based on current or lifetime income, and the results can differ greatly for some types of taxes.



QUESTIONS AND PROBLEMS

1. Why do most analysts assume that payroll taxes in the United States are borne by workers rather than by employers?
2. The demand for rutabagas is $Q = 2,000 - 300P$ and the supply of rutabagas is $Q = -100 + 100P$. Who bears the *statutory* incidence of a \$2 per unit tax on the sale of rutabagas? Who bears the *economic* incidence of this tax?
3. The demand for rutabagas is still $Q = 2,000 - 300P$ and the supply is still $Q = -100 + 100P$, as in Question 2. Governor Jackson decides that instead of imposing the \$2 sales tax described in Question 2, the government will instead force stores to pay the tax directly. What will happen to the “sticker price” on rutabagas? How will the size of the consumer tax burden change?
4. The demand for football tickets is $Q = 360 - 10P$ and the supply of football tickets is $Q = 20P$. Calculate the gross price paid by consumers after a per-ticket tax of \$4. Calculate the after-tax price received by ticket sellers.
5. The government is considering imposing taxes on the sellers of certain classes of products. The first tax they are considering is a tax on 2% milk. The second is a tax on all dairy products. The third is a tax on all food products. Which of these three taxes would you expect to have the largest impact on the sticker prices of the taxed products?
6. To finance a new health insurance program, the government of Millonia imposes a new \$2-per-hour payroll tax to be paid by employers.
 1. What do you expect to happen to wages and the size of the workforce?
 2. How will this answer change in markets where labor is inelastically demanded?
7. You have determined that producers, rather than consumers, will bear the lion’s share of the burden associated with a new tax. How does the elasticity of labor supply influence whether this tax burden will, in turn, be borne more by workers or more by property owners?
8. Why can some taxes that appear to be regressive in terms of current income be thought of as progressive from a lifetime tax incidence perspective?
9. Consider a labor market in which workers are paid the minimum wage. When will it matter for tax incidence whether a payroll tax is imposed on workers or on employers?
10. Consider the changes over time in the U.S. effective tax rates presented in [Table 19-1](#). How did the total effective tax burden change for the lowest and highest deciles of the population between 1979 and 2011? How did the composition of this burden across different types of taxes change over this period?

ADVANCED QUESTIONS

11. The elasticity of demand for maracas is -2.0 and the elasticity of supply is 3.0 . How much will the price of maracas change with a per-unit tax of \$2? Who bears the larger burden of the tax, consumers or producers?
12. The government of Byngia has introduced a new tax on tickets for airline travel. Byngia has two types of travelers, business travelers and leisure travelers. Business travelers in Byngia have an elasticity of demand of -1.2 , while Byngian leisure travelers have an elasticity of -3.0 . Airlines can price-discriminate among these groups; that is, they can charge different prices to the different types of fliers in the market. Which type of traveler will bear a larger fraction of the burden of the new tax on their tickets? Explain.
13. Massive Products, Inc., is a monopolist whose cost of production is given by $10Q + 2Q^2$ (so its marginal cost curve—equivalently, its inverse supply curve—is given by $10 + 4Q$). Demand for Massive Products' massive products is $Q = 200 - P$.
1. What price will the monopolist charge, and what profits will the monopolist earn? What will consumer surplus be?
 2. How will the monopolist's price and profits change if a tax of \$15 per unit is imposed on the buyers of the product?
 3. What is the deadweight burden of the tax?
14. Why did [Harding et al. \(2012\)](#), discussed in the text, use changes in state-level tax rates on cigarettes rather than across-state differences in tax rates on cigarettes in their study of the impact of cigarette taxes on cigarette prices? Based on their findings, would you expect consumers, on average, to bear a larger incidence of a new tax in a small state with many neighboring states—like Massachusetts—or in a large state like Texas? Explain.
15. The city of Malaise is considering a 10% tax on the revenues of all hotels/motels inside the city limits. Although not completely different from hotels and motels in the nearby suburbs, the ones in Malaise have a distinct advantage in their proximity to the interesting sights and convention centers. So individuals will pay some premium to stay in Malaise rather than to stay nearby.

Furthermore, all land is used equally well by hotels/motels and other forms of business; any Malaise land not taken by a hotel/motel is readily absorbed by other forms of business. Mayor Maladroit calls you in to advise him on the incidence of such a tax. He is particularly concerned with who will bear this tax in the short run (one month) and the long run (five years).

1. What is the incidence of the tax in the short run? Answer intuitively, and use a diagram to illustrate.
2. What is the long-run incidence? Once again, use a diagram to illustrate.
3. How would your analysis in part (b) change if hotels/motels in the suburbs were perfect substitutes for those in Malaise? What would happen to tax revenues?

APPENDIX TO CHAPTER 19 The Mathematics of Tax Incidence



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In this appendix, we develop the mathematical formulas that lie behind the incidence discussion in [Chapter 19](#). We then use them to examine various aspects of tax incidence in a monopoly market.

Tax Incidence Formulas

To analyze tax incidence, we need to measure how imposing a tax changes the price in a market. Consider first the case of a tax that is *paid by consumers* so that the total price change to consumers is

$$\text{total price change} = \Delta P + \tau$$

where ΔP is the change in market price (presumably negative), and τ is the tax payment. The price change to producers is just ΔP , the reduction in price.

We can determine the effect of tax on price in three steps:

Step 1:

Start with the definitions of elasticity of demand and supply:

$$\text{elasticity of demand} = \eta_d = \Delta Q / (\Delta P + \tau) \times (P/Q)$$

$$\text{elasticity of supply} = \eta_s = \Delta Q / \Delta P \times (P/Q)$$

Step 2:

Rearrange terms so that the two expressions can be set equal to each other:

$$\begin{aligned}\Delta Q/Q &= \eta_d \times (\Delta P + \tau)/P \\ &= \eta_s \times \Delta P/P\end{aligned}$$

Step 3:

Set these expressions equal to each other, and solve for the change in price as a function of the tax (which is what we need to know to do incidence analysis):

$$\Delta P = [\eta_d / (\eta_s - \eta_d)] \times \tau$$

This equation formalizes our intuition on inelastic and elastic factors. For example, if demand is inelastic ($\eta_d = 0$), then $\Delta P = 0$. The price does not change, so since the consumer is sending the check to the government, he bears the full tax. Alternatively, if demand is perfectly elastic ($\eta_d = \infty$), then $\Delta P = -\tau$. The price falls by the full amount of the tax, offsetting the consumer tax payment so that the consumer has no remaining tax burden.

Similarly, if we redo the exercise for a tax paid by producers, we obtain

$$\Delta P = [\eta_s / (\eta_s - \eta_d)] \times \tau$$

Tax Incidence in a Monopoly

We can put these mathematical insights to work to explore the implications of tax incidence in a monopoly. Suppose that a monopoly firm has a cost function of

$$C = 12 + q^2$$

Its marginal cost, the increment to cost from one additional unit of production, is therefore $2q$ (the derivative of the cost function with respect to quantity).

Suppose further that the monopolist faces a demand function of

$$p = 24 - q$$

Given the cost and demand functions, we can determine the monopolist's revenue and marginal revenue:

$$\text{revenues} = p \times q = (24 - q) \times q = 24q - q^2$$

$$\text{marginal revenues} = \text{derivative of revenues with respect to } q = 24 - 2q$$

Marginal revenues are set equal to marginal costs to maximize profits:

$$24 - 2q = 2q$$

Solving for q , we find that $q = 6$; marginal revenues equal marginal costs (and profits are maximized) by selling 6 units. Once the monopolist has determined the quantity at which profits are maximized, the price per unit is determined by the demand curve $p = 24 - 6 = 18$. Initially, then, the monopolist produces 6 units and sells them for \$18 each.

Now suppose that the government levies a tax on the monopolist of \$4 per unit sold. The cost function becomes $12 + q^2 + 4q$, and marginal costs are $2q + 4$. Re-solving for the profit-maximizing quantity, we obtain $q = 5$. As before, once the monopolist sets the quantity, the price is determined by the demand curve, $p = 24 - q = 19$. After the tax is imposed, the monopolist therefore produces 5 units at a price of \$19. The monopolist bears three-quarters of this tax; the price goes up by only \$1, but the monopolist pays \$4 in tax.

How would this outcome change if the government levies the tax on consumers of this monopoly product? After tax, the demand function becomes $p = 24 - q - 4 = 20 - q$. Revenues are therefore $(20 - q) \times q = 20q - q^2$. Differentiating with respect to q , we obtain a marginal revenue function of $20 - 2q$. Re-solving for the profit-maximizing quantity, we once again find that $q = 5$. On the new demand curve, this implies a price of \$15 to consumers and an after-tax cost to consumers of \$19 (because they pay the \$4 tax). So, once again, consumers bear only \$1 of the tax (since price has decreased from \$18 to \$15 to offset the \$4 tax payment), and the monopolist bears \$3 (as the price has decreased). This analysis

confirms our earlier finding that the side of the market on which the tax is imposed is irrelevant to the determination of which side bears the burden of the tax.

CHAPTER 20 Tax Inefficiencies and Their Implications for Optimal Taxation



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- [20.1 Taxation and Economic Efficiency](#)
- [20.2 Optimal Commodity Taxation](#)
- [20.3 Optimal Income Taxes](#)
- [20.4 Tax-Benefit Linkages and the Financing of Social Insurance Programs](#)
- [20.5 Conclusion](#)
- [APPENDIX TO CHAPTER 20](#)
 - [The Mathematics of Optimal Taxation](#)

